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A title with some math: $x = 1$

F. Irst,[a,b,1] **S. Econd,**[c] **T. Hird**[a,2] **and Fourth**[a,2]

[a]One University,
some-street, Country
[b]Another University,
different-address, Country
[c]A School for Advanced Studies,
some-location, Country

E-mail: first@one.univ, second@asas.edu, third@one.univ, fourth@one.univ

Abstract. Abstract...

[1]Corresponding author.
[2]Also at Some University.

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1 Gaussian covariance derivation

In general, the estimator for the power spectrum between fields α and β can be written as (here, ignoring the shot-noise contribution)

$$P_\ell^{\alpha\beta}(\boldsymbol{k}) = \frac{(2l+1)}{I_{22}^{\alpha\beta}} \int_{\hat{\boldsymbol{k}}} \frac{1}{2} \left(F_\ell^\alpha(\boldsymbol{k}) F_0^{\beta,*}(\boldsymbol{k}) + F_0^\alpha(\boldsymbol{k}) F_\ell^{\beta,*}(\boldsymbol{k}) \right), \tag{1.1}$$

where

$$F_\ell^\gamma(\boldsymbol{k}) = \int_{\boldsymbol{x}} W_{11}^\gamma(\boldsymbol{x}) e^{-i\boldsymbol{k}\cdot\boldsymbol{x}} \mathcal{L}_\ell(\hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{x}}) \gamma(\boldsymbol{x}) \tag{1.2}$$

and

$$I_{22}^{\alpha\beta} = \int_{\boldsymbol{x}} \bar{n}_\alpha(\boldsymbol{x}) \bar{n}_\beta(\boldsymbol{x}) w_\alpha(\boldsymbol{x}) w_\beta(\boldsymbol{x}) \tag{1.3}$$

for an arbitrary field γ and it has the following property

$$F_\ell^{\gamma,*}(-\boldsymbol{k}) = (-1)^\ell F_\ell^\gamma(\boldsymbol{k}) \tag{1.4}$$

since the field γ and its corresponding weights W_{11}^γ are real. The Gaussian covariance matrix of the power spectrum is given by

$$G_{\ell_1\ell_2}^{abcd}(k_1, k_2) = \langle P_{\ell_1}^{ab} P_{\ell_2}^{cd} \rangle - \langle P_{\ell_1}^{ab} \rangle \langle P_{\ell_2}^{cd} \rangle. \tag{1.5}$$

Substituting equation (1.1) into equation (1.5), we get

$$\begin{aligned} G_{\ell_1\ell_2}^{abcd}(k_1, k_2) = \frac{(2\ell_1+1)(2\ell_2+1)}{4 I_{22}^{ab} I_{22}^{cd}} \int_{\hat{\boldsymbol{k}}_1, \hat{\boldsymbol{k}}_2} \Big(\Big[& \left\langle F_{\ell_1}^a(\boldsymbol{k}_1) F_0^{b,*}(\boldsymbol{k}_1) F_{\ell_2}^c(\boldsymbol{k}_2) F_0^{d,*}(\boldsymbol{k}_2) \right\rangle \\ & - \langle F_{\ell_1}^a(\boldsymbol{k}_1) F_0^{b,*}(\boldsymbol{k}_1) \rangle \langle F_{\ell_2}^c(\boldsymbol{k}_2) F_0^{d,*}(\boldsymbol{k}_2) \rangle \Big] + \Big[\left\langle F_{\ell_1}^a(\boldsymbol{k}_1) F_0^{b,*}(\boldsymbol{k}_1) F_0^c(\boldsymbol{k}_2) F_{\ell_2}^{d,*}(\boldsymbol{k}_2) \right\rangle \\ & - \langle F_{\ell_1}^a(\boldsymbol{k}_1) F_0^{b,*}(\boldsymbol{k}_1) \rangle \langle F_0^c(\boldsymbol{k}_2) F_{\ell_2}^{d,*}(\boldsymbol{k}_2) \rangle \Big] + \Big[\left\langle F_0^a(\boldsymbol{k}_1) F_{\ell_1}^{b,*}(\boldsymbol{k}_1) F_{\ell_2}^c(\boldsymbol{k}_2) F_0^{d,*}(\boldsymbol{k}_2) \right\rangle \\ & - \langle F_0^a(\boldsymbol{k}_1) F_{\ell_1}^{b,*}(\boldsymbol{k}_1) \rangle \langle F_{\ell_2}^c(\boldsymbol{k}_2) F_0^{d,*}(\boldsymbol{k}_2) \rangle \Big] + \Big[\left\langle F_0^a(\boldsymbol{k}_1) F_{\ell_1}^{b,*}(\boldsymbol{k}_1) F_0^c(\boldsymbol{k}_2) F_{\ell_2}^{d,*}(\boldsymbol{k}_2) \right\rangle \\ & - \langle F_0^a(\boldsymbol{k}_1) F_{\ell_1}^{b,*}(\boldsymbol{k}_1) \rangle \langle F_0^c(\boldsymbol{k}_2) F_{\ell_2}^{d,*}(\boldsymbol{k}_2) \rangle \Big] \Big) \end{aligned} \tag{1.6}$$

We will focus on deriving the first term. The expressions for the other three terms can be inferred based on the first term using symmetry arguments. Applying Wick's theorem for

Gaussian random fields, for the first term, we get

$$\mathcal{G}^{abcd}_{\ell_1\ell_2}(\boldsymbol{k_1},\boldsymbol{k_2}) = \frac{(2\ell_1+1)(2\ell_2+1)}{4I^{ab}_{22}I^{cd}_{22}}\int_{\hat{\boldsymbol{k}}_1,\hat{\boldsymbol{k}}_2}\Big[\big\langle F^a_{\ell_1}(\boldsymbol{k}_1)\big(F^d_0(\boldsymbol{k}_2)\big)^*\big\rangle\big\langle F^c_{\ell_2}(\boldsymbol{k}_2)\big(F^b_0(\boldsymbol{k}_1)\big)^*\big\rangle + \big\langle F^a_{\ell_1}(\boldsymbol{k}_1)F^c_{\ell_2}(\boldsymbol{k}_2)\big\rangle\big\langle(F^b_0(\boldsymbol{k}_1))^\star\big(F^d_0(\boldsymbol{k}_2)\big)^*\big\rangle\Big]$$

$$= \frac{(2\ell_1+1)(2\ell_2+1)}{I^{ab}_{22}I^{cd}_{22}}\int_{\hat{\boldsymbol{k}}_1,\hat{\boldsymbol{k}}_2}\int_{\boldsymbol{x_1},\boldsymbol{x_1}',\boldsymbol{x_2},\boldsymbol{x_2}'} e^{-i\boldsymbol{k_1}\cdot(\boldsymbol{x_1}-\boldsymbol{x_1}')-i\boldsymbol{k_2}\cdot(\boldsymbol{x_2}-\boldsymbol{x_2}')}L_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}}_1)$$
$$\Big\{W^a_{11}(\boldsymbol{x_1})W^b_{11}(\boldsymbol{x_1}')W^c_{11}(\boldsymbol{x_2})W^d_{11}(\boldsymbol{x_2}')\langle a(\boldsymbol{x_1})d(\boldsymbol{x_2}')\rangle\langle c(\boldsymbol{x_2})b(\boldsymbol{x_1}')\rangle L_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_2)$$
$$+ W^a_{11}(\boldsymbol{x_1})W^b_{11}(\boldsymbol{x_1}')W^c_{11}(\boldsymbol{x_2}')W^d_{11}(\boldsymbol{x_2})\langle a(\boldsymbol{x_1})c(\boldsymbol{x_2}')\rangle\langle b(\boldsymbol{x_1}')d(\boldsymbol{x_2})\rangle L_{\ell_2}(-\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}'_2)\Big\} \tag{1.7}$$

For the second part of equation (1.7), we substitute $\boldsymbol{x_2}\leftrightarrow\boldsymbol{x_2}'$ and $\boldsymbol{k_2}\to-\boldsymbol{k_2}$ (These are variables are being integrated over, so we can swap them freely).

Substituting $\boldsymbol{s_1}=\boldsymbol{x_1}-\boldsymbol{x_2}'$ and $\boldsymbol{s_2}=\boldsymbol{x_2}-\boldsymbol{x_1}'$, we have

$$\mathcal{G}^{abcd}_{\ell_1\ell_2}(k_1,k_2) = \frac{(2\ell_1+1)(2\ell_2+1)}{4I^{ab}_{22}I^{cd}_{22}}\int_{\hat{\mathbf{k}}_1,\hat{\mathbf{k}}_2}\int_{\mathbf{x}_1,\mathbf{x}_2,\mathbf{s}_1,\mathbf{s}_2} e^{-i(\mathbf{k}_1-\mathbf{k}_2)\cdot(\mathbf{x}_1-\mathbf{x}_2)}e^{-i\mathbf{k}_1\cdot\mathbf{s}_2-i\mathbf{k}_2\cdot\mathbf{s}_1}$$
$$\times\Big[W^a_{11}(\mathbf{x}_1)W^d_{11}(\mathbf{x}_1-\mathbf{s}_1)W^c_{11}(\mathbf{x}_2)W^b_{11}(\mathbf{x}_2-\mathbf{s}_2)\mathcal{L}_{\ell_1}(\hat{\mathbf{x}}_1\cdot\hat{\mathbf{k}}_1)$$
$$\times\xi_{ad}(\mathbf{s}_1;\mathbf{x}_1)\xi_{bc}(\mathbf{s}_2;\mathbf{x}_2)\mathcal{L}_{\ell_2}(\hat{\mathbf{x}}_2\cdot\hat{\mathbf{k}}_2)+W^a_{11}(\mathbf{x}_1)W^c_{11}(\mathbf{x}_1-\mathbf{s}_1)W^d_{11}(\mathbf{x}_2)W^b_{11}(\mathbf{x}_2-\mathbf{s}_2)$$
$$\times\xi_{ac}(\mathbf{s}_1;\mathbf{x}_1)\xi_{bd}(\mathbf{s}_2;\mathbf{x}_2)\mathcal{L}_{\ell_1}(\hat{\mathbf{x}}_1\cdot\hat{\mathbf{k}}_1)(-1)^{\ell_2}\mathcal{L}_{\ell_2}(\hat{\mathbf{x}}_1\cdot\hat{\mathbf{k}}_2)]\Big].$$

In the last line, we assume the local plane parallel approximation such that $\boldsymbol{x_1}-\boldsymbol{s_1}=\boldsymbol{x_1}$. Integrating over $\boldsymbol{s_1}$ and $\boldsymbol{s_2}$, we will get

$$\mathcal{G}^{abcd}_{\ell_1\ell_2}(k_1,k_2)\simeq\sum_{\ell_1',\ell_2'}\Bigg\{\frac{(2\ell_1+1)(2\ell_2+1)}{4I^{ab}_{22}I^{cd}_{22}}\int_{\hat{\mathbf{k}}_1,\hat{\mathbf{k}}_2,\mathbf{x}_1,\mathbf{x}_2} e^{-i(\mathbf{x}_1-\mathbf{x}_2)\cdot(\mathbf{k}_1-\mathbf{k}_2)}$$
$$\times\mathcal{L}_{\ell_1}(\hat{\mathbf{x}}_1\cdot\hat{\mathbf{k}}_1)\mathcal{L}_{\ell_2'}(\hat{\mathbf{x}}_1\cdot\hat{\mathbf{k}}_2)\mathcal{L}_{\ell_1'}(\hat{\mathbf{x}}_2\cdot\hat{\mathbf{k}}_1)[W^{ad}_{22}(\mathbf{x}_1)W^{bc}_{22}(\mathbf{x}_2)P^{ad}_{\ell_2'}(k_2)P^{bc}_{\ell_1'}(k_1)\mathcal{L}_{\ell_2}(\hat{\mathbf{x}}_2\cdot\hat{\mathbf{k}}_2)$$
$$+W^{ac}_{22}(\mathbf{x}_1)W^{bd}_{22}(\mathbf{x}_2)P^{ac}_{\ell_2'}(k_2)P^{bd}_{\ell_1'}(k_1)(-1)^{\ell_2}\mathcal{L}_{\ell_2}(\hat{\mathbf{x}}_1\cdot\hat{\mathbf{k}}_2)]\Bigg\},$$
$$\equiv\sum_{\ell_1',\ell_2'}P^{ad}_{\ell_2'}(k_2)P^{bc}_{\ell_1'}(k_1)\mathcal{W}^{(ad,bc)}_{\ell_1,\ell_2,\ell_1',\ell_2'}(k_1,k_2,x_2,x_1)+$$
$$\sum_{\ell_1',\ell_2'}P^{ac}_{\ell_2'}(k_2)P^{bd}_{\ell_1'}(k_1)(-1)^{\ell_2}\mathcal{W}^{(ac,bd)}_{\ell_1,\ell_2,\ell_1',\ell_2'}(k_1,k_2,x_1,x_1) \tag{1.8}$$

where

$$\mathcal{W}^{(\alpha\beta,\gamma\epsilon)}_{\ell_1,\ell_2,\ell_1',\ell_2'}(k_1,k_2,x_\zeta,x_\eta) = \frac{(2\ell_1+1)(2\ell_2+1)}{4I^{ab}_{22}I^{cd}_{22}}\int_{\hat{\mathbf{k}}_1,\hat{\mathbf{k}}_2,\mathbf{x}_1,\mathbf{x}_2} e^{-i(\mathbf{x}_1-\mathbf{x}_2)\cdot(\mathbf{k}_1-\mathbf{k}_2)}$$
$$\times\mathcal{L}_{\ell_1}(\hat{\mathbf{x}}_\eta\cdot\hat{\mathbf{k}}_1)\mathcal{L}_{\ell_1'}(\hat{\mathbf{x}}_2\cdot\hat{\mathbf{k}}_1)\mathcal{L}_{\ell_2'}(\hat{\mathbf{x}}_1\cdot\hat{\mathbf{k}}_2)W^{\alpha\epsilon}_{22}(\mathbf{x}_1)W^{\beta\gamma}_{22}(\mathbf{x}_2)\mathcal{L}_{\ell_2}(\hat{\mathbf{x}}_\zeta\cdot\hat{\mathbf{k}}_2) \tag{1.9}$$

Using symmetry in equations (1.6), the full expression for the Gaussian covariance matrix is

$$
\begin{aligned}
G^{abcd}_{\ell_1\ell_2}(k_1,k_2) = &\sum_{\ell_1',\ell_2'} P^{ad}_{\ell_2'}(k_2)P^{bc}_{\ell_1'}(k_1)\mathcal{W}^{(ad,bc)}_{\ell_1,\ell_2,\ell_1',\ell_2'}(k_1,k_2,x_2,x_1)+\\
&\sum_{\ell_1',\ell_2'} P^{ac}_{\ell_2'}(k_2)P^{bd}_{\ell_1'}(k_1)(-1)^{\ell_2}\mathcal{W}^{(ac,bd)}_{\ell_1,\ell_2,\ell_1',\ell_2'}(k_1,k_2,x_1,x_1)\\
&+\sum_{\ell_1',\ell_2'} P^{ad}_{\ell_2'}(k_2)P^{bc}_{\ell_1'}(k_1)\mathcal{W}^{(ad,bc)}_{\ell_1,\ell_2,\ell_1',\ell_2'}(k_1,k_2,x_1,x_1)+\\
&\sum_{\ell_1',\ell_2'} P^{ac}_{\ell_2'}(k_2)P^{bd}_{\ell_1'}(k_1)(-1)^{\ell_2}\mathcal{W}^{(ac,bd)}_{\ell_1,\ell_2,\ell_1',\ell_2'}(k_1,k_2,x_2,x_1)\\
&\sum_{\ell_1',\ell_2'} P^{ad}_{\ell_2'}(k_2)P^{bc}_{\ell_1'}(k_1)\mathcal{W}^{(ad,bc)}_{\ell_1,\ell_2,\ell_1',\ell_2'}(k_1,k_2,x_2,x_2)+\\
&\sum_{\ell_1',\ell_2'} P^{ac}_{\ell_2'}(k_2)P^{bd}_{\ell_1'}(k_1)(-1)^{\ell_2}\mathcal{W}^{(ac,bd)}_{\ell_1,\ell_2,\ell_1',\ell_2'}(k_1,k_2,x_1,x_2)\\
&\sum_{\ell_1',\ell_2'} P^{ad}_{\ell_2'}(k_2)P^{bc}_{\ell_1'}(k_1)\mathcal{W}^{(ad,bc)}_{\ell_1,\ell_2,\ell_1',\ell_2'}(k_1,k_2,x_1,x_2)+\\
&\sum_{\ell_1',\ell_2'} P^{ac}_{\ell_2'}(k_2)P^{bd}_{\ell_1'}(k_1)(-1)^{\ell_2}\mathcal{W}^{(ac,bd)}_{\ell_1,\ell_2,\ell_1',\ell_2'}(k_1,k_2,x_2,x_2) \qquad (1.10)
\end{aligned}
$$

For the auto-covariance matrix of the galaxy power spectrum multipole, $a = b = c = d = g$ and ℓ_1, ℓ_2 are even. Equation (1.10) reduces to

$$
\begin{aligned}
G^{abcd}_{\ell_1\ell_2}(k_1,k_2) = &\sum_{\ell_1',\ell_2'} P^{gg}_{\ell_2'}(k_2)P^{gg}_{\ell_1'}(k_1)\frac{(2\ell_1+1)(2\ell_2+1)}{4I^{gg}_{22}I^{gg}_{22}}\int_{\hat{\mathbf{k}}_1,\hat{\mathbf{k}}_2,\mathbf{x}_1,\mathbf{x}_2} e^{-i(\mathbf{x}_1-\mathbf{x}_2)\cdot(\mathbf{k}_1-\mathbf{k}_2)}W^{gg}_{22}(x_1)W^{gg}_{22}(x_2)\\
&\mathcal{L}_{\ell_1'}(\hat{\mathbf{x}}_2\cdot\hat{\mathbf{k}}_1)\mathcal{L}_{\ell_2'}(\hat{\mathbf{x}}_1\cdot\hat{\mathbf{k}}_2)\Big[\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2)+\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2)\\
&+\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2)+\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2)\\
&+\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2)+\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2)\\
&+\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2)+\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2)\Big]\\
= &\sum_{\ell_1',\ell_2'} P^{gg}_{\ell_2'}(k_2)P^{gg}_{\ell_1'}(k_1)\frac{(2\ell_1+1)(2\ell_2+1)}{2I^{gg}_{22}I^{gg}_{22}}\int_{\hat{\mathbf{k}}_1,\hat{\mathbf{k}}_2,\mathbf{x}_1,\mathbf{x}_2} e^{-i(\mathbf{x}_1-\mathbf{x}_2)\cdot(\mathbf{k}_1-\mathbf{k}_2)}W^{gg}_{22}(x_1)W^{gg}_{22}(x_2)\\
&\mathcal{L}_{\ell_1'}(\hat{\mathbf{x}}_2\cdot\hat{\mathbf{k}}_1)\mathcal{L}_{\ell_2'}(\hat{\mathbf{x}}_1\cdot\hat{\mathbf{k}}_2)\Big[\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2)+\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2)\\
&+\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2)+\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2)\Big]\\
= &\sum_{\ell_1',\ell_2'} P^{gg}_{\ell_2'}(k_2)P^{gg}_{\ell_1'}(k_1)\frac{(2\ell_1+1)(2\ell_2+1)}{I^{gg}_{22}I^{gg}_{22}}\int_{\hat{\mathbf{k}}_1,\hat{\mathbf{k}}_2,\mathbf{x}_1,\mathbf{x}_2} e^{-i(\mathbf{x}_1-\mathbf{x}_2)\cdot(\mathbf{k}_1-\mathbf{k}_2)}W^{gg}_{22}(x_1)W^{gg}_{22}(x_2)\\
&\mathcal{L}_{\ell_1'}(\hat{\mathbf{x}}_2\cdot\hat{\mathbf{k}}_1)\mathcal{L}_{\ell_2'}(\hat{\mathbf{x}}_1\cdot\hat{\mathbf{k}}_2)\Big[\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2)+\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2)\Big] \qquad (1.11)
\end{aligned}
$$

consistent with ref. [1]. In the third equality, we swap $x_1 \leftrightarrow x_2$, $k_1 \to -k_1$ and $k_2 \to -k_2$ for the third and fourth terms since they are variables being integrated over.

2 Shot-noise for the Gaussian covariance matrix

Following Appendix B of ref. [1], the discrete analogue of the momentum field estimator should be

$$F_\ell^p(\boldsymbol{k}) = \sum_i^{N_p} w_i^p(\boldsymbol{x}) u_i(\boldsymbol{x}) e^{-i\boldsymbol{k}\cdot\boldsymbol{x}} L(\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{x}}) \tag{2.1}$$

using the definition from ref. [2]. Notice that N_p denotes the number of galaxies with peculiar velocity measurements and w^p is the FKP weight for the momentum field [2]

$$w^p(\boldsymbol{x}) = \frac{1}{u^2(\boldsymbol{x}) + \bar{n}_p(\boldsymbol{x}) P^p(\boldsymbol{k},\boldsymbol{x})}. \tag{2.2}$$

The auto-power spectrum of the momentum can be represented as

$$\begin{aligned}
\langle F_{\ell_1}^p(\boldsymbol{k}_1) F_{\ell_2}^p(\boldsymbol{k}_2)\rangle &= \langle \sum_i^{N_g}\sum_j^{N_g} w_i^p w_j^p u_i u_j e^{-i\boldsymbol{k}_1\cdot\boldsymbol{x}_i - i\boldsymbol{k}_2\cdot\boldsymbol{x}_j} \mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}}_i)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_j)\rangle \\
&= \langle \sum_i^{N_g}\sum_{j\neq i}^{N_g} w_i^p w_j^p u_i u_j e^{-i\boldsymbol{k}_1\cdot\boldsymbol{x}_i - i\boldsymbol{k}_2\cdot\boldsymbol{x}_j} \mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}}_i)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_j)\rangle \\
&+ \langle \sum_i^{N_g} w_i^{p2} u_i^2 e^{-i(\boldsymbol{k}_1+\boldsymbol{k}_2)\cdot\boldsymbol{x}_i} \mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}}_i)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_i)\rangle \\
&\simeq \int_{\boldsymbol{x}_1,\boldsymbol{x}_2} W_{11}^p(\boldsymbol{x}_1) W_{11}^p(\boldsymbol{x}_2) \langle (1+\delta(\boldsymbol{x}_1)) u(\boldsymbol{x}_1) (1+\delta(\boldsymbol{x}_2)) u(\boldsymbol{x}_2)\rangle e^{-i\boldsymbol{k}_1\cdot\boldsymbol{x}_1 - i\boldsymbol{k}_2\cdot\boldsymbol{x}_2} \\
&\times \mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_2) \\
&+ \int_{\boldsymbol{x}} W_{12}^p(\boldsymbol{x}) \langle u^2(\boldsymbol{x})\rangle e^{-i(\boldsymbol{k}_1+\boldsymbol{k}_2)\cdot\boldsymbol{x}} \mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}})\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}) \\
&= \langle F_{\ell_1}^p(\boldsymbol{k}_1) F_{\ell_2}^p(\boldsymbol{k}_2)\rangle_{\rm cont} + \mathcal{I}_{12}^{p\ \ell_1,\ell_2}(\boldsymbol{k}_1,\boldsymbol{k}_2)
\end{aligned} \tag{2.3, 2.4}$$

The continuous integral is

$$\begin{aligned}
\langle F_{\ell_1}^p(\boldsymbol{k}_1) F_{\ell_2}^p(\boldsymbol{k}_2)\rangle_{\rm cont} &= \int_{\boldsymbol{x}_1,\boldsymbol{x}_2} W_{11}^p(\boldsymbol{x}_1) W_{11}^p(\boldsymbol{x}_2) \langle (1+\delta(\boldsymbol{x}_1)) u(\boldsymbol{x}_1) (1+\delta(\boldsymbol{x}_2)) u(\boldsymbol{x}_2)\rangle e^{-i\boldsymbol{k}_1\cdot\boldsymbol{x}_1 - i\boldsymbol{k}_2\cdot\boldsymbol{x}_2} \\
&\times \mathcal{L}_{\ell_1}(\boldsymbol{k}_1\cdot\boldsymbol{x}_1)\mathcal{L}_{\ell_2}(\boldsymbol{k}_2\cdot\boldsymbol{x}_2)
\end{aligned} \tag{2.5}$$

and the shot noise contribution of the momentum auto-power spectrum is defined as

$$\mathcal{I}_{12}^{p\ \ell_1,\ell_2}(\boldsymbol{k}_1,\boldsymbol{k}_2) \equiv \int_{\boldsymbol{x}} W_{12}^p(\boldsymbol{x}) \langle u^2(\boldsymbol{x})\rangle e^{-i(\boldsymbol{k}_1+\boldsymbol{k}_2)\cdot\boldsymbol{x}} \mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}})\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}) \tag{2.6}$$

We further calculate the cross-power spectrum between galaxy density and momentum first

$$\begin{aligned}\langle F^d_{\ell_1}(\boldsymbol{k_1})F^p_{\ell_2}(\boldsymbol{k_2})\rangle &= \langle(\sum_i^{N_g} -\alpha\sum_i^{N_r})\sum_j^{N_p} w_i^g w_j^p u(\boldsymbol{x}_j)e^{-i\boldsymbol{k}_1\cdot\boldsymbol{x}_i - i\boldsymbol{k}_2\cdot\boldsymbol{x}_j}\mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}_1}\cdot\hat{\boldsymbol{x}}_i)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}_2}\cdot\hat{\boldsymbol{x}}_j)\rangle \\ &= \langle(\sum_i^{N_g} -\alpha\sum_i^{N_r})\sum_{j\neq i}^{N_p} w_i^g w_j^p u(\boldsymbol{x}_j)e^{-i\boldsymbol{k}_1\cdot\boldsymbol{x}_i - i\boldsymbol{k}_2\cdot\boldsymbol{x}_j}\mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}_1}\cdot\hat{\boldsymbol{x}}_i)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}_2}\cdot\hat{\boldsymbol{x}}_j)\rangle \\ &+ \langle\sum_i^{N_p} w_i^g w_i^p u(\boldsymbol{x}_i)e^{-i(\boldsymbol{k}_1+\boldsymbol{k}_2)\cdot\boldsymbol{x}_i}\mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}_1}\cdot\hat{\boldsymbol{x}}_i)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}_2}\cdot\hat{\boldsymbol{x}}_i)\rangle. \end{aligned} \tag{2.7}$$

For the shot-noise contribution in the second equality, since the random catalogue does not overlap with either the galaxy catalogue or the peculiar velocity catalogue, the cross term between the peculiar velocity catalogue and the random catalogue is zero. Secondly, the only overlap between the galaxy catalogue and the peculiar velocity catalogue is galaxies that have peculiar velocity measurements. Therefore, we sum up to N_p instead of N_g.[1] Focusing on the shot-noise term, in the continuous limit, it should be

$$\int_{\boldsymbol{x}} \bar{n}_p(\boldsymbol{x})w^g(\boldsymbol{x})w^p(\boldsymbol{x})\langle u(\boldsymbol{x})\rangle e^{-i(\boldsymbol{k}_1+\boldsymbol{k}_2)\cdot\boldsymbol{x}}\mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}_1}\cdot\hat{\boldsymbol{x}})\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}_2}\cdot\hat{\boldsymbol{x}}). \tag{2.8}$$

Based on the cosmological principle, $\langle u(\boldsymbol{x})\rangle = 0$, so the cross-power spectrum has no shot-noise contribution, consistent with ref. [2]. Consequently, the galaxy-momentum cross-power spectrum in the continuous limit is

$$\langle F^d_{\ell_1}(\boldsymbol{k_1})F^p_{\ell_2}(\boldsymbol{k_2})\rangle = \int_{\boldsymbol{x}_1,\boldsymbol{x_2}} W^g_{11}(\boldsymbol{x}_1)W^p_{11}(\boldsymbol{x}_2)\langle\delta(\boldsymbol{x}_1)(1+\delta(\boldsymbol{x}_2))u(\boldsymbol{x}_2)\rangle e^{-i\boldsymbol{k}_1\cdot\boldsymbol{x}_1 - i\boldsymbol{k}_2\cdot\boldsymbol{x}_2}\mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}_1}\cdot\hat{\boldsymbol{x}_1})\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}_2}\cdot\hat{\boldsymbol{x}_2})\rangle. \tag{2.9}$$

From symmetry, the momentum galaxy cross-power spectrum is given by

$$\langle F^p_{\ell_1}(\boldsymbol{k_1})F^d_{\ell_2}(\boldsymbol{k_2})\rangle = \int_{\boldsymbol{x}_1,\boldsymbol{x_2}} W^g_{11}(\boldsymbol{x}_2)W^p_{11}(\boldsymbol{x}_1)\langle(1+\delta(\boldsymbol{x}_1))u(\boldsymbol{x}_1)\delta(\boldsymbol{x}_2)\rangle e^{-i\boldsymbol{k}_1\cdot\boldsymbol{x}_1 - i\boldsymbol{k}_2\cdot\boldsymbol{x}_2}\mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}_1}\cdot\hat{\boldsymbol{x}_1})\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}_2}\cdot\hat{\boldsymbol{x}_2})\rangle. \tag{2.10}$$

pp-pp Following Eq. B4 of [1], the short-noise contribution of the Gaussian covariance of the momentum auto-power spectrum is

$$\begin{aligned} C^{pp-pp,\mathrm{SN-G}}_{\ell_1\ell_2}(k_1,k_2) = \frac{(2\ell_1+1)(2\ell_2+1)}{I^{pp\,2}_{22}}\int_{\hat{\mathbf{k}}_1,\hat{\mathbf{k}}_2}\Big[&\langle F^p_{\ell_1}(\mathbf{k}_1)F^p_{\ell_2}(-\mathbf{k}_2)\rangle\mathcal{I}^{p\ 0,0}_{12}(-\mathbf{k}_1,\mathbf{k}_2) \\ &+ \langle F^p_0(\mathbf{k}_1)F^0_0(-\mathbf{k}_2)\rangle\mathcal{I}^{p\ \ell_1,\ell_2}_{12}(-\mathbf{k}_1,\mathbf{k}_2) + \langle F^p_{\ell_1}(\mathbf{k}_1)F^p_0(-\mathbf{k}_2)\rangle\mathcal{I}^{p\ 0,\ell_2}_{12}(-\mathbf{k}_1,\mathbf{k}_2) \\ &+ \langle F^p_0(\mathbf{k}_1)F^p_{\ell_2}(-\mathbf{k}_2)\rangle\mathcal{I}^{p\ \ell_1,0}_{12}(-\mathbf{k}_1,\mathbf{k}_2) + \mathcal{I}^{p\ \ell_1,0}_{12}(\mathbf{k}_1,-\mathbf{k}_2)\mathcal{I}^{p\ 0,\ell_2}_{12}(-\mathbf{k}_1,\mathbf{k}_2) \\ &+ \mathcal{I}^{p\ \ell_1,\ell_2}_{12}(\mathbf{k}_1,-\mathbf{k}_2)\mathcal{I}^{p\ 0,0}_{12}(-\mathbf{k}_1,\mathbf{k}_2)\Big]. \end{aligned} \tag{2.11}$$

[1] This is what has been done in the DESI PV survey, peculiar velocity measurements outside the BGS footprint are not included in the final analysis. Therefore, there will be no peculiar velocity measurements outside the BGS galaxy sample.

$$\int_{\hat{\mathbf{k}}_1,\hat{\mathbf{k}}_2} \mathcal{I}_{12}^{p\ \ell_1,\ell_2}(\mathbf{k}_1,-\mathbf{k}_2)\,\mathcal{I}_{12}^{p\ 0,0}(-\mathbf{k}_1,\mathbf{k}_2)+\mathcal{I}_{12}^{p\ \ell_1,0}(\mathbf{k}_1,-\mathbf{k}_2)\,\mathcal{I}_{12}^{p\ 0,\ell_2}(-\mathbf{k}_1,\mathbf{k}_2)$$
$$=\int_{\hat{\mathbf{k}}_1,\hat{\mathbf{k}}_2,\mathbf{x}_1,\mathbf{x}_2} W_{12}^p(\mathbf{x}_1)\,W_{12}^p(\mathbf{x}_2)\,\langle u^2(\mathbf{x}_1)\rangle\langle u^2(\mathbf{x}_2)\rangle e^{-i(\mathbf{k}_1-\mathbf{k}_2)\cdot(\mathbf{x}_1-\mathbf{x}_2)}\mathcal{L}_{\ell_1}(\hat{\mathbf{x}}_1\cdot\hat{\mathbf{k}}_1)\Big[\mathcal{L}_{\ell_2}(\hat{\mathbf{x}}_1\cdot\hat{\mathbf{k}}_2)+\mathcal{L}_{\ell_2}(\hat{\mathbf{x}}_2\cdot\hat{\mathbf{k}}_2)\Big]. \tag{2.12}$$

under endpoint-LOS approximation

$$\langle F_{\ell_1}^p(\boldsymbol{k}_1)F_{\ell_2}^p(-\boldsymbol{k}_2)\rangle_{\rm cont}=\int_{\boldsymbol{x}_1,\boldsymbol{x}_1'} W_{11}^p(\boldsymbol{x}_1)W_{11}^p(\boldsymbol{x}_1')\langle(1+\delta(\boldsymbol{x}_1))u(\boldsymbol{x}_1)(1+\delta(\boldsymbol{x}_1'))u(\boldsymbol{x}_1')\rangle e^{-i\boldsymbol{k}_1\cdot\boldsymbol{x}_1+i\boldsymbol{k}_2\cdot\boldsymbol{x}_1'}$$
$$\times\,\mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_1)$$

substitute $\mathbf{x}_1'$ with $\mathbf{x}_1-\mathbf{s}_1$

$$\begin{aligned}\langle F_{\ell_1}^p(\boldsymbol{k}_1)\,F_{\ell_2}^p(-\boldsymbol{k}_2)\rangle_{\rm cont}&=\int_{\boldsymbol{x}_1,\boldsymbol{s}_1} W_{11}^p(\boldsymbol{x}_1)\,W_{11}^p(\boldsymbol{x}_1-\boldsymbol{s}_1)\,\xi_{pp}(\boldsymbol{s}_1;\boldsymbol{x}_1)\\&\quad\times e^{-i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{x}_1}\,e^{-i\boldsymbol{k}_2\cdot\boldsymbol{s}_1}\,\mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}}_1)\,\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_1)\\&\simeq\int_{\boldsymbol{x}_1} e^{-i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{x}_1}\,P_{pp}^{\rm local}(\boldsymbol{k}_2;\boldsymbol{x}_1)\,W_{22}(\boldsymbol{x}_1)\,\mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}}_1)\,\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_1)\\&=\sum_{\ell'}P_{\ell'}^{pp}(k_2)\int_{\boldsymbol{x}_1} e^{-i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{x}_1}\,W_{22}(\boldsymbol{x}_1)\,\mathcal{L}_{\ell'}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_1)\,\mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}}_1)\,\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_1)\end{aligned} \tag{2.13}$$

To restore the symmetry of $\{\boldsymbol{k}_1,\ell_1\}-\{\boldsymbol{k}_2,\ell_2\}$, the above equation can be written as

$$\begin{aligned}\langle F_{\ell_1}^p(\boldsymbol{k}_1)\,F_{\ell_2}^p(-\boldsymbol{k}_2)\rangle_{\rm cont}&\simeq\int_{\boldsymbol{x}_1} e^{-i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{x}_1}\,W_{22}(\boldsymbol{x}_1)\,\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\,\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2)\\&\quad\sum_{\ell'}\frac{1}{2}\Big[P_{\ell'}^{pp}(k_1)\,\mathcal{L}_{\ell'}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)+P_{\ell'}^{pp}(k_2)\,\mathcal{L}_{\ell'}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2)\Big]\end{aligned} \tag{2.14}$$

$$\begin{aligned}&C_{\ell_1\ell_2}^{pp-pp,{\rm SN-G}}(k_1,k_2)=\frac{(2\ell_1+1)(2\ell_2+1)}{I_{22}^2}\int_{\hat{\boldsymbol{k}}_1,\hat{\boldsymbol{k}}_2,\boldsymbol{x}_1,\boldsymbol{x}_2}\\&\Big\{W_{12}(\boldsymbol{x}_1)\,W_{12}(\boldsymbol{x}_2)\,\langle u^2(\boldsymbol{x}_1)\rangle\langle u^2(\boldsymbol{x}_2)\rangle e^{-i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot(\boldsymbol{x}_1-\boldsymbol{x}_2)}\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\Big[\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2)+\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2)\Big]\\&+e^{-i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot(\boldsymbol{x}_1-\boldsymbol{x}_2)}W_{22}(\boldsymbol{x}_1)\,W_{12}(\boldsymbol{x}_2)\langle u^2(\boldsymbol{x}_2)\rangle\left[\frac{1}{2}\sum_{\ell'}P_{\ell'}^{pp}(k_1)\,\mathcal{L}_{\ell'}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)+\frac{1}{2}\sum_{\ell'}P_{\ell'}^{pp}(k_2)\,\mathcal{L}_{\ell'}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2)\right]\\&\times\Big[\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2)+\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2)+\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2)+\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2)\Big]\Big\}.\end{aligned} \tag{2.15}$$

$$C^{pp-pp,\mathrm{SN-G}}_{\ell_1\ell_2}(k_1,k_2) = \left[\sum_{\ell'} P^{pp}_{\ell'}(k_1)\,\mathcal{W}^{(2-1)}_{\ell_1,\ell_2,\ell'}(k_1,k_2) + P^{pp}_{\ell'}(k_2)\,\mathcal{W}^{(2-2)}_{\ell_1,\ell_2,\ell'}(k_1,k_2)\right] + \mathcal{W}^{(3)}_{\ell_1,\ell_2}(k_1,k_2) \tag{2.16}$$

$$\mathcal{W}^{(2-1)}_{\ell_1,\ell_2,\ell'}(k_1,k_2) \equiv \frac{(2\ell_1+1)(2\ell_2+1)}{2{I^p_{22}}^2}\int_{\hat{\boldsymbol{k}}_1,\hat{\boldsymbol{k}}_2,\boldsymbol{x}_1,\boldsymbol{x}_2} e^{-i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot(\boldsymbol{x}_1-\boldsymbol{x}_2)} W_{22}(\boldsymbol{x}_1)W_{12}(\boldsymbol{x}_2)\langle u^2(\boldsymbol{x}_2)\rangle \mathcal{L}_{\ell'}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)$$
$$\Big\{\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2) + \mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2) + \mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2) + \mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2)\Big\}. \tag{2.17}$$

$$\mathcal{W}^{(2-2)}_{\ell_1,\ell_2,\ell'}(k_1,k_2) \equiv \frac{(2\ell_1+1)(2\ell_2+1)}{2{I^p_{22}}^2}\int_{\hat{\boldsymbol{k}}_1,\hat{\boldsymbol{k}}_2,\boldsymbol{x}_1,\boldsymbol{x}_2} e^{-i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot(\boldsymbol{x}_1-\boldsymbol{x}_2)} W_{22}(\boldsymbol{x}_1)W_{12}(\boldsymbol{x}_2)\langle u^2(\boldsymbol{x}_2)\rangle \mathcal{L}_{\ell'}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2)$$
$$\Big\{\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2) + \mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2) + \mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2) + \mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_1)\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2)\Big\}. \tag{2.18}$$

$$\mathcal{W}^{(3)}_{\ell_1,\ell_2}(k_1,k_2) \equiv \frac{(2\ell_1+1)(2\ell_2+1)}{{I^p_{22}}^2}\int_{\hat{\boldsymbol{k}}_1,\hat{\boldsymbol{k}}_2,\boldsymbol{x}_1,\boldsymbol{x}_2} W_{12}(\boldsymbol{x}_1)W_{12}(\boldsymbol{x}_2)\langle u^2(\boldsymbol{x}_1)\rangle\langle u^2(\boldsymbol{x}_2)\rangle$$
$$\times\, e^{-i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot(\boldsymbol{x}_1-\boldsymbol{x}_2)}\mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\Big[\mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2) + \mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_2\cdot\hat{\boldsymbol{k}}_2)\Big]. \tag{2.19}$$

2.1 General form of shot-noise

$$\hat{P}^{ab}_{\ell_1}(k_1) = \frac{2\ell_1+1}{2I^{ab}_{22}}\int_{\hat{k}_1}\Big[F^a_{\ell_1}(\mathbf{k}_1)F^{b,*}_0(\mathbf{k}_1) + F^a_0(\mathbf{k}_1)F^{b,*}_{\ell_1}(\mathbf{k}_1)\Big], \tag{2.20}$$

$$\hat{P}^{cd}_{\ell_2}(k_2) = \frac{2\ell_2+1}{2I^{cd}_{22}}\int_{\hat{k}_2}\Big[F^c_{\ell_2}(\mathbf{k}_2)F^{d,*}_0(\mathbf{k}_2) + F^c_0(\mathbf{k}_2)F^{d,*}_{\ell_2}(\mathbf{k}_2)\Big]. \tag{2.21}$$

$$
\begin{aligned}
C_{\ell_1\ell_2}^{ab-cd,\mathrm{SN}} = \frac{(2\ell_1+1)(2\ell_2+1)}{4I_{22}^{ab}I_{22}^{cd}} \int_{\hat{k}_1,\hat{k}_2} \Big[& \\
& \mathcal{I}^{ac}_{\ell_1,\ell_2}(\mathbf{k}_1,\mathbf{k}_2) \left\langle F_0^b(-\mathbf{k}_1) F_0^d(-\mathbf{k}_2) \right\rangle_{\mathrm{cont}} + \left\langle F^a_{\ell_1}(\mathbf{k}_1) F^c_{\ell_2}(\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \mathcal{I}^{bd}_{0,0}(-\mathbf{k}_1,-\mathbf{k}_2) \\
& + \mathcal{I}^{ac}_{\ell_1,\ell_2}(\mathbf{k}_1,\mathbf{k}_2) \mathcal{I}^{bd}_{0,0}(-\mathbf{k}_1,-\mathbf{k}_2) \\
& + \mathcal{I}^{ad}_{\ell_1,0}(\mathbf{k}_1,-\mathbf{k}_2) \left\langle F_0^b(-\mathbf{k}_1) F^c_{\ell_2}(\mathbf{k}_2) \right\rangle_{\mathrm{cont}} + \left\langle F^a_{\ell_1}(\mathbf{k}_1) F_0^d(-\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \mathcal{I}^{bc}_{0,\ell_2}(-\mathbf{k}_1,\mathbf{k}_2) \\
& + \mathcal{I}^{ad}_{\ell_1,0}(\mathbf{k}_1,-\mathbf{k}_2) \mathcal{I}^{bc}_{0,\ell_2}(-\mathbf{k}_1,\mathbf{k}_2) \textit{Abovecomesfrom} \langle F^a_{\ell_1}(\mathbf{k}_1) F_0^{b,*}(\mathbf{k}_1) F^c_{\ell_2}(\mathbf{k}_2) F_0^{d,*}(\mathbf{k}_2) \rangle \\
& + (-1)^{\ell_2} \mathcal{I}^{ac}_{\ell_1,0}(\mathbf{k}_1,\mathbf{k}_2) \left\langle F_0^b(-\mathbf{k}_1) F^d_{\ell_2}(-\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \\
& + (-1)^{\ell_2} \left\langle F^a_{\ell_1}(\mathbf{k}_1) F_0^c(\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \mathcal{I}^{bd}_{0,\ell_2}(-\mathbf{k}_1,-\mathbf{k}_2) + (-1)^{\ell_2} \mathcal{I}^{ac}_{\ell_1,0}(\mathbf{k}_1,\mathbf{k}_2) \mathcal{I}^{bd}_{0,\ell_2}(-\mathbf{k}_1,-\mathbf{k}_2) \\
& + (-1)^{\ell_2} \mathcal{I}^{ad}_{\ell_1,\ell_2}(\mathbf{k}_1,-\mathbf{k}_2) \left\langle F_0^b(-\mathbf{k}_1) F_0^c(\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \\
& + (-1)^{\ell_2} \left\langle F^a_{\ell_1}(\mathbf{k}_1) F^d_{\ell_2}(-\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \mathcal{I}^{bc}_{0,0}(-\mathbf{k}_1,\mathbf{k}_2) + (-1)^{\ell_2} \mathcal{I}^{ad}_{\ell_1,\ell_2}(\mathbf{k}_1,-\mathbf{k}_2) \mathcal{I}^{bc}_{0,0}(-\mathbf{k}_1,\mathbf{k}_2) \\
& \textit{Abovecomesfrom} \langle F^a_{\ell_1}(\mathbf{k}_1) F_0^{b,*}(\mathbf{k}_1) F_0^c(\mathbf{k}_2) F^{d,*}_{\ell_2}(\mathbf{k}_2) \rangle \\
& + (-1)^{\ell_1} \mathcal{I}^{ac}_{0,\ell_2}(\mathbf{k}_1,\mathbf{k}_2) \left\langle F^b_{\ell_1}(-\mathbf{k}_1) F_0^d(-\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \\
& + (-1)^{\ell_1} \left\langle F_0^a(\mathbf{k}_1) F_0^c(\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \mathcal{I}^{bd}_{\ell_1,0}(-\mathbf{k}_1,-\mathbf{k}_2) + (-1)^{\ell_1} \mathcal{I}^{ac}_{0,\ell_2}(\mathbf{k}_1,\mathbf{k}_2) \mathcal{I}^{bd}_{\ell_1,0}(-\mathbf{k}_1,-\mathbf{k}_2) \\
& + (-1)^{\ell_1} \mathcal{I}^{ad}_{0,0}(\mathbf{k}_1,-\mathbf{k}_2) \left\langle F^b_{\ell_1}(-\mathbf{k}_1) F^c_{\ell_2}(\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \\
& + (-1)^{\ell_1} \left\langle F_0^a(\mathbf{k}_1) F_0^d(-\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \mathcal{I}^{bc}_{\ell_1,\ell_2}(-\mathbf{k}_1,\mathbf{k}_2) + (-1)^{\ell_1} \mathcal{I}^{ad}_{0,0}(\mathbf{k}_1,-\mathbf{k}_2) \mathcal{I}^{bc}_{\ell_1,\ell_2}(-\mathbf{k}_1,\mathbf{k}_2) \\
& \textit{Abovecomesfrom} \langle F^a_0(\mathbf{k}_1) F^{b,*}_{\ell_1}(\mathbf{k}_1) F^c_{\ell_2}(\mathbf{k}_2) F_0^{d,*}(\mathbf{k}_2) \rangle \\
& + (-1)^{\ell_1+\ell_2} \mathcal{I}^{ac}_{0,0}(\mathbf{k}_1,\mathbf{k}_2) \left\langle F^b_{\ell_1}(-\mathbf{k}_1) F^d_{\ell_2}(-\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \\
& + (-1)^{\ell_1+\ell_2} \left\langle F_0^a(\mathbf{k}_1) F_0^c(\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \mathcal{I}^{bd}_{\ell_1,\ell_2}(-\mathbf{k}_1,-\mathbf{k}_2) + (-1)^{\ell_1+\ell_2} \mathcal{I}^{ac}_{0,0}(\mathbf{k}_1,\mathbf{k}_2) \mathcal{I}^{bd}_{\ell_1,\ell_2}(-\mathbf{k}_1,-\mathbf{k}_2) \\
& + (-1)^{\ell_1+\ell_2} \mathcal{I}^{ad}_{0,\ell_2}(\mathbf{k}_1,-\mathbf{k}_2) \left\langle F^b_{\ell_1}(-\mathbf{k}_1) F_0^c(\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \\
& + (-1)^{\ell_1+\ell_2} \left\langle F_0^a(\mathbf{k}_1) F^d_{\ell_2}(-\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \mathcal{I}^{bc}_{\ell_1,0}(-\mathbf{k}_1,\mathbf{k}_2) \\
& + (-1)^{\ell_1+\ell_2} \mathcal{I}^{ad}_{0,\ell_2}(\mathbf{k}_1,-\mathbf{k}_2) \mathcal{I}^{bc}_{\ell_1,0}(-\mathbf{k}_1,\mathbf{k}_2) \Big]. \\
& \textit{Abovecomesfrom} \langle F^a_0(\mathbf{k}_1) F^{b,*}_{\ell_1}(\mathbf{k}_1) F_0^c(\mathbf{k}_2) F^{d,*}_{\ell_2}(\mathbf{k}_2) \rangle
\end{aligned}
\tag{2.22}
$$

dd-dd

$$
\begin{aligned}
C_{\ell_1\ell_2}^{dd-dd,\mathrm{SN}}(k_1,k_2) = \frac{(2\ell_1+1)(2\ell_2+1)}{(I_{22}^{dd})^2} \int_{\hat{k}_1,\hat{k}_2} \Big[& \\
& \left\langle F^d_{\ell_1}(\mathbf{k}_1) F^d_{\ell_2}(-\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \mathcal{I}^{dd}_{0,0}(-\mathbf{k}_1,\mathbf{k}_2) + \left\langle F_0^d(\mathbf{k}_1) F_0^d(-\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \mathcal{I}^{dd}_{\ell_1,\ell_2}(-\mathbf{k}_1,\mathbf{k}_2) \\
& + \left\langle F^d_{\ell_1}(\mathbf{k}_1) F_0^d(-\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \mathcal{I}^{dd}_{0,\ell_2}(-\mathbf{k}_1,\mathbf{k}_2) + \left\langle F_0^d(\mathbf{k}_1) F^d_{\ell_2}(-\mathbf{k}_2) \right\rangle_{\mathrm{cont}} \mathcal{I}^{dd}_{\ell_1,0}(-\mathbf{k}_1,\mathbf{k}_2) \\
& + \mathcal{I}^{dd}_{\ell_1,0}(\mathbf{k}_1,-\mathbf{k}_2) \mathcal{I}^{dd}_{0,\ell_2}(-\mathbf{k}_1,\mathbf{k}_2) + \mathcal{I}^{dd}_{\ell_1,\ell_2}(\mathbf{k}_1,-\mathbf{k}_2) \mathcal{I}^{dd}_{0,0}(-\mathbf{k}_1,\mathbf{k}_2) \Big].
\end{aligned}
\tag{2.23}
$$

pp-pp

$$C_{\ell_1\ell_2}^{pp-pp,\mathrm{SN}}(k_1,k_2) = \frac{(2\ell_1+1)(2\ell_2+1)}{(I_{22}^{pp})^2}\int_{\hat{k}_1,\hat{k}_2}\Big[\\ \Big\langle F_{\ell_1}^p(\mathbf{k}_1)F_{\ell_2}^p(-\mathbf{k}_2)\Big\rangle_{\mathrm{cont}}\mathcal{I}_{0,0}^{pp}(-\mathbf{k}_1,\mathbf{k}_2) + \langle F_0^p(\mathbf{k}_1)F_0^p(-\mathbf{k}_2)\rangle_{\mathrm{cont}}\mathcal{I}_{\ell_1,\ell_2}^{pp}(-\mathbf{k}_1,\mathbf{k}_2) \\ + \Big\langle F_{\ell_1}^p(\mathbf{k}_1)F_0^p(-\mathbf{k}_2)\Big\rangle_{\mathrm{cont}}\mathcal{I}_{0,\ell_2}^{pp}(-\mathbf{k}_1,\mathbf{k}_2) + \Big\langle F_0^p(\mathbf{k}_1)F_{\ell_2}^p(-\mathbf{k}_2)\Big\rangle_{\mathrm{cont}}\mathcal{I}_{\ell_1,0}^{pp}(-\mathbf{k}_1,\mathbf{k}_2) \\ + \mathcal{I}_{\ell_1,0}^{pp}(\mathbf{k}_1,-\mathbf{k}_2)\mathcal{I}_{0,\ell_2}^{pp}(-\mathbf{k}_1,\mathbf{k}_2) + \mathcal{I}_{\ell_1,\ell_2}^{pp}(\mathbf{k}_1,-\mathbf{k}_2)\mathcal{I}_{0,0}^{pp}(-\mathbf{k}_1,\mathbf{k}_2)\Big]. \qquad (2.24)$$

pd-pd

$$C_{\ell_1\ell_2}^{pd-pd,\mathrm{SN}} = \frac{(2\ell_1+1)(2\ell_2+1)}{4I_{22}^{pd}I_{22}^{pd}}\int_{\hat{k}_1,\hat{k}_2}\Big[\\ \mathcal{I}_{\ell_1,\ell_2}^{p}(\mathbf{k}_1,\mathbf{k}_2)\Big\langle F_0^{d,*}(\mathbf{k}_1)F_0^{d,*}(\mathbf{k}_2)\Big\rangle_{\mathrm{cont}} + \Big\langle F_{\ell_1}^p(\mathbf{k}_1)F_{\ell_2}^p(\mathbf{k}_2)\Big\rangle_{\mathrm{cont}}\mathcal{I}_{0,0}^{d}(-\mathbf{k}_1,-\mathbf{k}_2) \\ + \mathcal{I}_{\ell_1,\ell_2}^{p}(\mathbf{k}_1,\mathbf{k}_2)\mathcal{I}_{0,0}^{d}(-\mathbf{k}_1,-\mathbf{k}_2) \\ + \mathcal{I}_{\ell_1,0}^{p}(\mathbf{k}_1,\mathbf{k}_2)\Big\langle F_0^{d,*}(\mathbf{k}_1)F_{\ell_2}^{d,*}(\mathbf{k}_2)\Big\rangle_{\mathrm{cont}} + (-1)^{\ell_2}\Big\langle F_{\ell_1}^p(\mathbf{k}_1)F_0^p(\mathbf{k}_2)\Big\rangle_{\mathrm{cont}}\mathcal{I}_{0,\ell_2}^{d}(-\mathbf{k}_1,-\mathbf{k}_2) \\ + (-1)^{\ell_2}\mathcal{I}_{\ell_1,0}^{p}(\mathbf{k}_1,\mathbf{k}_2)\mathcal{I}_{0,\ell_2}^{d}(-\mathbf{k}_1,-\mathbf{k}_2) \\ + \mathcal{I}_{0,\ell_2}^{p}(\mathbf{k}_1,\mathbf{k}_2)\Big\langle F_{\ell_1}^{d,*}(\mathbf{k}_1)F_0^{d,*}(\mathbf{k}_2)\Big\rangle_{\mathrm{cont}} + (-1)^{\ell_1}\Big\langle F_0^p(\mathbf{k}_1)F_{\ell_2}^p(\mathbf{k}_2)\Big\rangle_{\mathrm{cont}}\mathcal{I}_{\ell_1,0}^{d}(-\mathbf{k}_1,-\mathbf{k}_2) \\ + (-1)^{\ell_1}\mathcal{I}_{0,\ell_2}^{p}(\mathbf{k}_1,\mathbf{k}_2)\mathcal{I}_{\ell_1,0}^{d}(-\mathbf{k}_1,-\mathbf{k}_2) \\ + \mathcal{I}_{0,0}^{p}(\mathbf{k}_1,\mathbf{k}_2)\Big\langle F_{\ell_1}^{d,*}(\mathbf{k}_1)F_{\ell_2}^{d,*}(\mathbf{k}_2)\Big\rangle_{\mathrm{cont}} + (-1)^{\ell_1+\ell_2}\langle F_0^p(\mathbf{k}_1)F_0^p(\mathbf{k}_2)\rangle_{\mathrm{cont}}\mathcal{I}_{\ell_1,\ell_2}^{d}(-\mathbf{k}_1,-\mathbf{k}_2) \\ + (-1)^{\ell_1+\ell_2}\mathcal{I}_{0,0}^{p}(\mathbf{k}_1,\mathbf{k}_2)\mathcal{I}_{\ell_1,\ell_2}^{d}(-\mathbf{k}_1,-\mathbf{k}_2)\Big]. \qquad (2.25)$$

Due to the symmetry argument (galaxy number density power spectrum sign does not depend on the observation angle, but the cross power spectrum between galaxy number density and the peculiar velocity does), for the *pd* cross spectrum, only odd multipoles contribute.

$$C_{\ell_1\ell_2}^{pd-pd,\mathrm{SN}} = \frac{(2\ell_1+1)(2\ell_2+1)}{4I_{22}^{pd}I_{22}^{pd}}\int_{\hat{k}_1,\hat{k}_2}\Big[\\ \mathcal{I}_{\ell_1,\ell_2}^{p}(\mathbf{k}_1,\mathbf{k}_2)\Big\langle F_0^{d,*}(\mathbf{k}_1)F_0^{d,*}(\mathbf{k}_2)\Big\rangle_{\mathrm{cont}} + \Big\langle F_{\ell_1}^p(\mathbf{k}_1)F_{\ell_2}^p(\mathbf{k}_2)\Big\rangle_{\mathrm{cont}}\mathcal{I}_{0,0}^{d}(-\mathbf{k}_1,-\mathbf{k}_2) \\ \quad + \mathcal{I}_{\ell_1,\ell_2}^{p}(\mathbf{k}_1,\mathbf{k}_2)\mathcal{I}_{0,0}^{d}(-\mathbf{k}_1,-\mathbf{k}_2) \\ + \mathcal{I}_{\ell_1,0}^{p}(\mathbf{k}_1,\mathbf{k}_2)\Big\langle F_0^{d,*}(\mathbf{k}_1)F_{\ell_2}^{d,*}(\mathbf{k}_2)\Big\rangle_{\mathrm{cont}} - \Big\langle F_{\ell_1}^p(\mathbf{k}_1)F_0^p(\mathbf{k}_2)\Big\rangle_{\mathrm{cont}}\mathcal{I}_{0,\ell_2}^{d}(-\mathbf{k}_1,-\mathbf{k}_2) \\ \quad - \mathcal{I}_{\ell_1,0}^{p}(\mathbf{k}_1,\mathbf{k}_2)\mathcal{I}_{0,\ell_2}^{d}(-\mathbf{k}_1,-\mathbf{k}_2) \\ + \mathcal{I}_{0,\ell_2}^{p}(\mathbf{k}_1,\mathbf{k}_2)\Big\langle F_{\ell_1}^{d,*}(\mathbf{k}_1)F_0^{d,*}(\mathbf{k}_2)\Big\rangle_{\mathrm{cont}} - \Big\langle F_0^p(\mathbf{k}_1)F_{\ell_2}^p(\mathbf{k}_2)\Big\rangle_{\mathrm{cont}}\mathcal{I}_{\ell_1,0}^{d}(-\mathbf{k}_1,-\mathbf{k}_2) \\ \quad - \mathcal{I}_{0,\ell_2}^{p}(\mathbf{k}_1,\mathbf{k}_2)\mathcal{I}_{\ell_1,0}^{d}(-\mathbf{k}_1,-\mathbf{k}_2) \\ + \mathcal{I}_{0,0}^{p}(\mathbf{k}_1,\mathbf{k}_2)\Big\langle F_{\ell_1}^{d,*}(\mathbf{k}_1)F_{\ell_2}^{d,*}(\mathbf{k}_2)\Big\rangle_{\mathrm{cont}} + \langle F_0^p(\mathbf{k}_1)F_0^p(\mathbf{k}_2)\rangle_{\mathrm{cont}}\mathcal{I}_{\ell_1,\ell_2}^{d}(-\mathbf{k}_1,-\mathbf{k}_2) \\ \quad + \mathcal{I}_{0,0}^{p}(\mathbf{k}_1,\mathbf{k}_2)\mathcal{I}_{\ell_1,\ell_2}^{d}(-\mathbf{k}_1,-\mathbf{k}_2)\Big]. \qquad (2.26)$$

$$
\begin{aligned}
&\int_{\hat{k}_1,\hat{k}_2}\Big[\mathcal{I}^p_{\ell_1,\ell_2}(\mathbf{k}_1,\mathbf{k}_2)\mathcal{I}^d_{0,0}(-\mathbf{k}_1,-\mathbf{k}_2)-\mathcal{I}^p_{\ell_1,0}(\mathbf{k}_1,\mathbf{k}_2)\mathcal{I}^d_{0,\ell_2}(-\mathbf{k}_1,-\mathbf{k}_2)\\
&-\mathcal{I}^p_{0,\ell_2}(\mathbf{k}_1,\mathbf{k}_2)\mathcal{I}^d_{\ell_1,0}(-\mathbf{k}_1,-\mathbf{k}_2)+\mathcal{I}^p_{0,0}(\mathbf{k}_1,\mathbf{k}_2)\mathcal{I}^d_{\ell_1,\ell_2}(-\mathbf{k}_1,-\mathbf{k}_2)\Big]\\
&=(1+\bar{\alpha})\int_{\hat{k}_1,\hat{k}_2}\int d^3x_1\,d^3x_2\,W^p_{12}(\mathbf{x}_1)\left\langle u^2(\mathbf{x}_1)\right\rangle W^d_{12}(\mathbf{x}_2)\times e^{-i(\mathbf{k}_1+\mathbf{k}_2)\cdot(\mathbf{x}_1-\mathbf{x}_2)}\\
&\quad\times\Big[\mathcal{L}_{\ell_1}(\hat{\mathbf{k}}_1\cdot\hat{\mathbf{x}}_1)\mathcal{L}_{\ell_2}(\hat{\mathbf{k}}_2\cdot\hat{\mathbf{x}}_1)-(-1)^{\ell_2}\mathcal{L}_{\ell_1}(\hat{\mathbf{k}}_1\cdot\hat{\mathbf{x}}_1)\mathcal{L}_{\ell_2}(\hat{\mathbf{k}}_2\cdot\hat{\mathbf{x}}_2)\\
&\qquad-(-1)^{\ell_1}\mathcal{L}_{\ell_1}(\hat{\mathbf{k}}_1\cdot\hat{\mathbf{x}}_2)\mathcal{L}_{\ell_2}(\hat{\mathbf{k}}_2\cdot\hat{\mathbf{x}}_1)+(-1)^{\ell_1+\ell_2}\mathcal{L}_{\ell_1}(\hat{\mathbf{k}}_1\cdot\hat{\mathbf{x}}_2)\mathcal{L}_{\ell_2}(\hat{\mathbf{k}}_2\cdot\hat{\mathbf{x}}_2)\Big].
\end{aligned}
\tag{2.27}
$$

pd-pp

$$
\begin{aligned}
C^{pd-pp,\mathrm{SN}}_{\ell_1\ell_2}=\frac{(2\ell_1+1)(2\ell_2+1)}{4I^{pd}_{22}I^{pp}_{22}}\int_{\hat{k}_1,\hat{k}_2}\Big[&\\
&\mathcal{I}^{pp}_{\ell_1,\ell_2}(\mathbf{k}_1,\mathbf{k}_2)\left\langle F^{d,*}_0(\mathbf{k}_1)F^{p,*}_0(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}+\mathcal{I}^{pp}_{\ell_1,0}(\mathbf{k}_1,-\mathbf{k}_2)\left\langle F^{d,*}_0(\mathbf{k}_1)F^{p}_{\ell_2}(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}\\
&+\mathcal{I}^{pp}_{\ell_1,0}(\mathbf{k}_1,\mathbf{k}_2)\left\langle F^{d,*}_0(\mathbf{k}_1)F^{p,*}_{\ell_2}(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}+\mathcal{I}^{pp}_{\ell_1,\ell_2}(\mathbf{k}_1,-\mathbf{k}_2)\left\langle F^{d,*}_0(\mathbf{k}_1)F^{p}_{0}(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}\\
&+\mathcal{I}^{pp}_{0,\ell_2}(\mathbf{k}_1,\mathbf{k}_2)\left\langle F^{d,*}_{\ell_1}(\mathbf{k}_1)F^{p,*}_0(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}+\mathcal{I}^{pp}_{0,0}(\mathbf{k}_1,-\mathbf{k}_2)\left\langle F^{d,*}_{\ell_1}(\mathbf{k}_1)F^{p}_{\ell_2}(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}\\
&+\mathcal{I}^{pp}_{0,0}(\mathbf{k}_1,\mathbf{k}_2)\left\langle F^{d,*}_{\ell_1}(\mathbf{k}_1)F^{p,*}_{\ell_2}(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}+\mathcal{I}^{pp}_{0,\ell_2}(\mathbf{k}_1,-\mathbf{k}_2)\left\langle F^{d,*}_{\ell_1}(\mathbf{k}_1)F^{p}_{0}(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}\Big].
\end{aligned}
\tag{2.28}
$$

pd-dd

$$
\begin{aligned}
C^{pd-dd,\mathrm{SN}}_{\ell_1\ell_2}=\frac{(2\ell_1+1)(2\ell_2+1)}{4I^{pd}_{22}I^{dd}_{22}}\int_{\hat{k}_1,\hat{k}_2}\Big[&\\
&\left\langle F^{p}_{\ell_1}(\mathbf{k}_1)F^{d}_{\ell_2}(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}\mathcal{I}^d_{0,0}(-\mathbf{k}_1,-\mathbf{k}_2)+\left\langle F^{p}_{\ell_1}(\mathbf{k}_1)F^{d,*}_{0}(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}\mathcal{I}^d_{0,\ell_2}(-\mathbf{k}_1,\mathbf{k}_2)\\
&+\left\langle F^{p}_{\ell_1}(\mathbf{k}_1)F^{d}_{0}(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}\mathcal{I}^d_{0,\ell_2}(-\mathbf{k}_1,-\mathbf{k}_2)+\left\langle F^{p}_{\ell_1}(\mathbf{k}_1)F^{d,*}_{\ell_2}(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}\mathcal{I}^d_{0,0}(-\mathbf{k}_1,\mathbf{k}_2)\\
&-\left\langle F^{p}_{0}(\mathbf{k}_1)F^{d}_{\ell_2}(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}\mathcal{I}^d_{\ell_1,0}(-\mathbf{k}_1,-\mathbf{k}_2)-\left\langle F^{p}_{0}(\mathbf{k}_1)F^{d,*}_{0}(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}\mathcal{I}^d_{\ell_1,\ell_2}(-\mathbf{k}_1,\mathbf{k}_2)\\
&-\left\langle F^{p}_{0}(\mathbf{k}_1)F^{d}_{0}(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}\mathcal{I}^d_{\ell_1,\ell_2}(-\mathbf{k}_1,-\mathbf{k}_2)-\left\langle F^{p}_{0}(\mathbf{k}_1)F^{d,*}_{\ell_2}(\mathbf{k}_2)\right\rangle_{\mathrm{cont}}\mathcal{I}^d_{\ell_1,0}(-\mathbf{k}_1,\mathbf{k}_2)\Big].
\end{aligned}
\tag{2.29}
$$

pp-dd There's no shot-noise.

$$
\begin{aligned}
\langle F^p_{\ell_1}(\boldsymbol{k}_1)\,F^d_{\ell_2}(-\boldsymbol{k}_2)\rangle_{\mathrm{cont}}&=\int_{\boldsymbol{x}_1,\boldsymbol{s}_1}W^p_{11}(\boldsymbol{x}_1)\,W^d_{11}(\boldsymbol{x}_1-\boldsymbol{s}_1)\,\xi_{pd}(\boldsymbol{s}_1;\boldsymbol{x}_1)\\
&\quad\times e^{-i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{x}_1}\,e^{-i\boldsymbol{k}_2\cdot\boldsymbol{s}_1}\,\mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}}_1)\,\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_1)\\
&\simeq\int_{\boldsymbol{x}_1}e^{-i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{x}_1}\,P^{\mathrm{local}}_{pd}(\boldsymbol{k}_2;\boldsymbol{x}_1)\,W^p_{11}(\boldsymbol{x}_1)\,W^d_{11}(\boldsymbol{x}_1)\,\mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}}_1)\,\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_1)\\
&=\sum_{\ell'}P^{pd}_{\ell'}(k_2)\int_{\boldsymbol{x}_1}e^{-i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{x}_1}\,W^{pd}_{22}(\boldsymbol{x}_1)\,\mathcal{L}_{\ell'}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_1)\,\mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}}_1)\,\mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_1)
\end{aligned}
\tag{2.30}
$$

$$\langle F^p_{\ell_1}(\boldsymbol{k}_1)\, F^d_{\ell_2}(-\boldsymbol{k}_2)\rangle_{\rm cont} \simeq \int_{\boldsymbol{x}_1} e^{-i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{x}_1}\, W^p_{11}(\boldsymbol{x}_1)\, W^d_{11}(\boldsymbol{x}_1)\, \mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\, \mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2) \sum_{\ell'} \frac{1}{2}\left[P^{pd}_{\ell'}(k_1)\, \mathcal{L}_{\ell'}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1) + P^{pd}_{\ell'}(k_2)\, \mathcal{L}_{\ell'}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2)\right] \tag{2.31}$$

$$\begin{aligned}\langle F^p_{\ell_1}(\boldsymbol{k}_1)\, F^d_{\ell_2}(\boldsymbol{k}_2)\rangle_{\rm cont} &= \int_{\boldsymbol{x}_1,\boldsymbol{s}_1} W^p_{11}(\boldsymbol{x}_1)\, W^d_{11}(\boldsymbol{x}_1-\boldsymbol{s}_1)\, \xi_{pd}(\boldsymbol{s}_1;\boldsymbol{x}_1) \\ &\quad \times e^{-i(\boldsymbol{k}_1+\boldsymbol{k}_2)\cdot\boldsymbol{x}_1}\, e^{i\boldsymbol{k}_2\cdot\boldsymbol{s}_1}\, \mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}}_1)\, \mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_1) \\ &\simeq \int_{\boldsymbol{x}_1} e^{-i(\boldsymbol{k}_1+\boldsymbol{k}_2)\cdot\boldsymbol{x}_1}\, P^{\rm local}_{pd}(-\boldsymbol{k}_2;\boldsymbol{x}_1)\, W^p_{11}(\boldsymbol{x}_1)\, W^d_{11}(\boldsymbol{x}_1)\, \mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}}_1)\, \mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_1) \\ &= \sum_{\ell'} P^{pd}_{\ell'}(k_2)(-1)^{\ell'} \int_{\boldsymbol{x}_1} e^{-i(\boldsymbol{k}_1+\boldsymbol{k}_2)\cdot\boldsymbol{x}_1}\, W^{pd}_{22}(\boldsymbol{x}_1)\, \mathcal{L}_{\ell'}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_1)\, \mathcal{L}_{\ell_1}(\hat{\boldsymbol{k}}_1\cdot\hat{\boldsymbol{x}}_1)\, \mathcal{L}_{\ell_2}(\hat{\boldsymbol{k}}_2\cdot\hat{\boldsymbol{x}}_1)\end{aligned} \tag{2.32}$$

$$\langle F^p_{\ell_1}(\boldsymbol{k}_1)\, F^d_{\ell_2}(\boldsymbol{k}_2)\rangle_{\rm cont} \simeq \int_{\boldsymbol{x}_1} e^{-i(\boldsymbol{k}_1+\boldsymbol{k}_2)\cdot\boldsymbol{x}_1}\, W^{pd}_{22}(\boldsymbol{x}_1)\, \mathcal{L}_{\ell_1}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1)\, \mathcal{L}_{\ell_2}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2) \sum_{\ell'} \frac{1}{2}(-1)^{\ell'}\left[P^{pd}_{\ell'}(k_1)\, \mathcal{L}_{\ell'}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_1) + P^{pd}_{\ell'}(k_2)\, \mathcal{L}_{\ell'}(\hat{\boldsymbol{x}}_1\cdot\hat{\boldsymbol{k}}_2)\right] \tag{2.33}$$

A Some title

Please always give a title also for appendices.

Acknowledgments

This is the most common positions for acknowledgments. A macro is available to maintain the same layout and spelling of the heading.

Note added. This is also a good position for notes added after the paper has been written.

References

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